

Fuzzy Logic

Understanding Fuzzy Logic

- Fuzzy logic provides a framework for reasoning that is approximate rather than exact.
- Unlike traditional binary logic (0 or 1), fuzzy logic allows for varying degrees of truth.
- Enables more human-like decision-making.
- Example: In temperature sensing, a traditional system classifies temperatures as “cold” or “warm.”
- Fuzzy logic system can describe temperatures as “slightly warm” or “fairly cold.”
- Used in AI, automation, and decision-making systems.

Perspectives & Key Takeaways

- **Two Perspectives of Fuzzy Logic:**
 - Broadly: A technique for handling reasoning that is not strictly binary.
 - Specifically: Generalizes multivalued logics studied in symbolic logic.
- **Importance:**
 - Enhances human-like reasoning.
 - Reduces rigid classifications.
 - Crucial in AI systems and control mechanisms.

Classical (Crisp) Sets

- Classical (crisp) sets have strict boundaries where an element either belongs or does not.
- Defined by a **universe of discourse**, a collection of elements sharing common characteristics.
- **Examples:**
 - Set of boys and girls in a particular age group.
 - Clock speeds of computer CPUs.
 - Integers greater than 8: $A = \{x \mid x > 8\}$
- **Boundary is fixed**

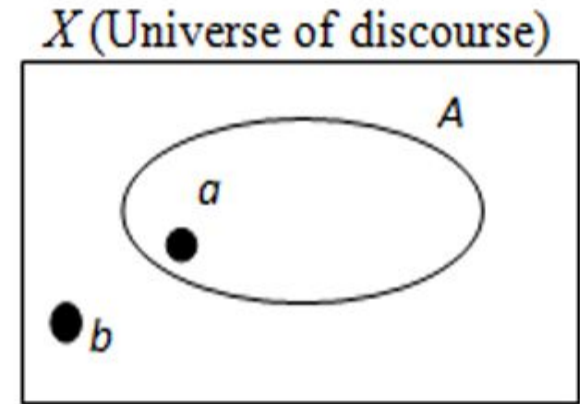


Fig.4.1: Crisp set A

Limitations of Classical Sets

- Classical sets can lead to **unrealistic classifications** due to rigid boundaries.
- **Example:**
 - A “tall person” is defined as someone over 6 feet.
 - A person who is **5.999 feet** is not considered tall, which is unreasonable.
- **Human concepts are fluid:**
 - Someone feeling cold at **7°C** will likely feel similarly at **6.999°C**.
 - Classical sets **fail to capture** this nuance.
- **Solution:**
 - Fuzzy sets allow for **gradual membership**, better representing real-world scenarios.

Fuzzy Sets

- Unlike crisp sets, fuzzy sets do not have sharp boundaries.
- The boundary is ambiguous, allowing partial membership.
- **Membership values:**
 - A clear member of the set has a value of **1**.
 - A non-member has a value of **0**.
 - An element on the boundary has a membership value between **0 and 1**.
- The transition from “belonging” to “not belonging” is smooth.
- **Membership functions** define the degree of membership and provide flexibility for modeling real-world scenarios.

X (Universe of discourse)

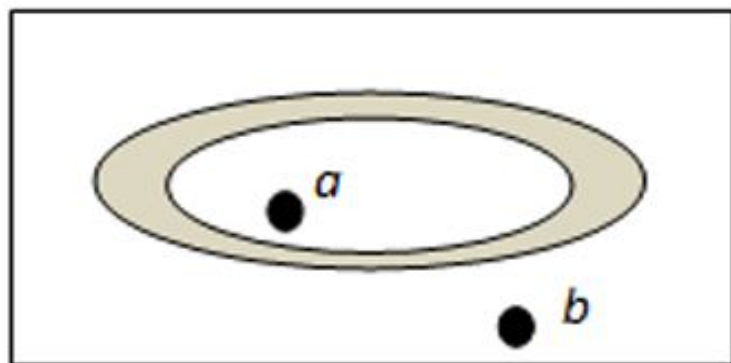


Fig.4.2: Fuzzy set A

Example of Fuzzy Sets

- **Comparison: Crisp vs. Fuzzy Set**
 - **Crisp Set:** A person taller than 6 feet is “tall.”
 - **Fuzzy Set:** Height is a gradual measure—someone slightly below 6 feet may still be considered tall.
- **Universe of discourse:** Heights from **3 feet to 9 feet**.
- The term “**tall**” is represented as a curve that defines the degree of tallness.
- In abstract concepts like numbers, sharp boundaries may make sense.
- However, real-world concepts, such as human height, require a more flexible classification.
- Fuzzy sets allow for **gradual inclusion**, avoiding unnatural distinctions.

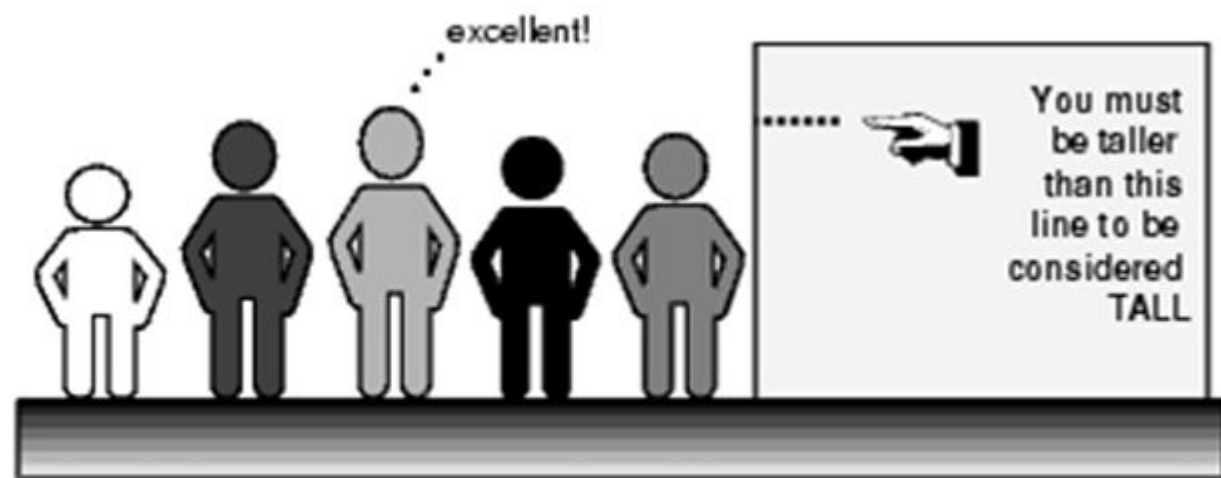
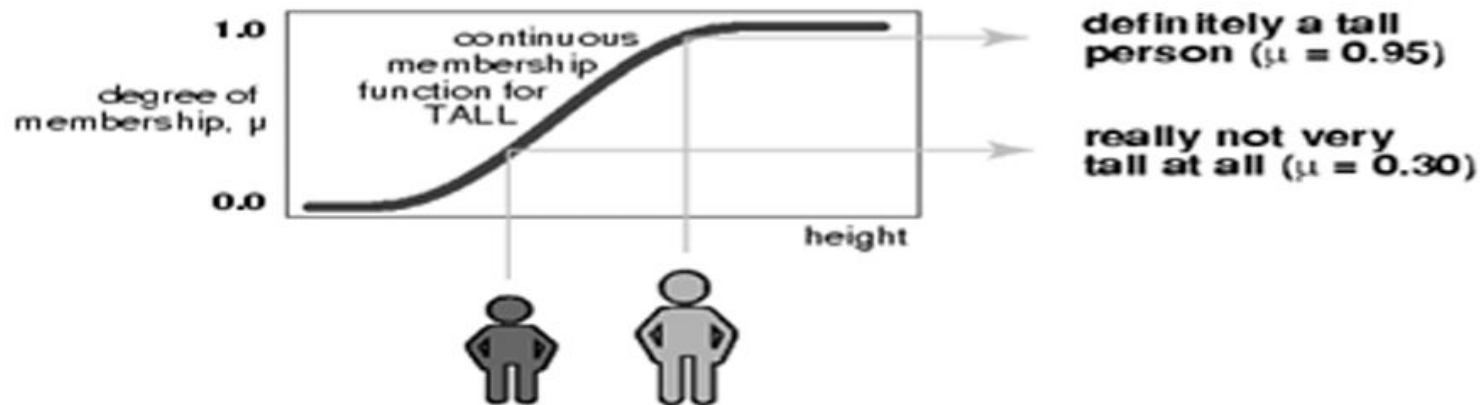
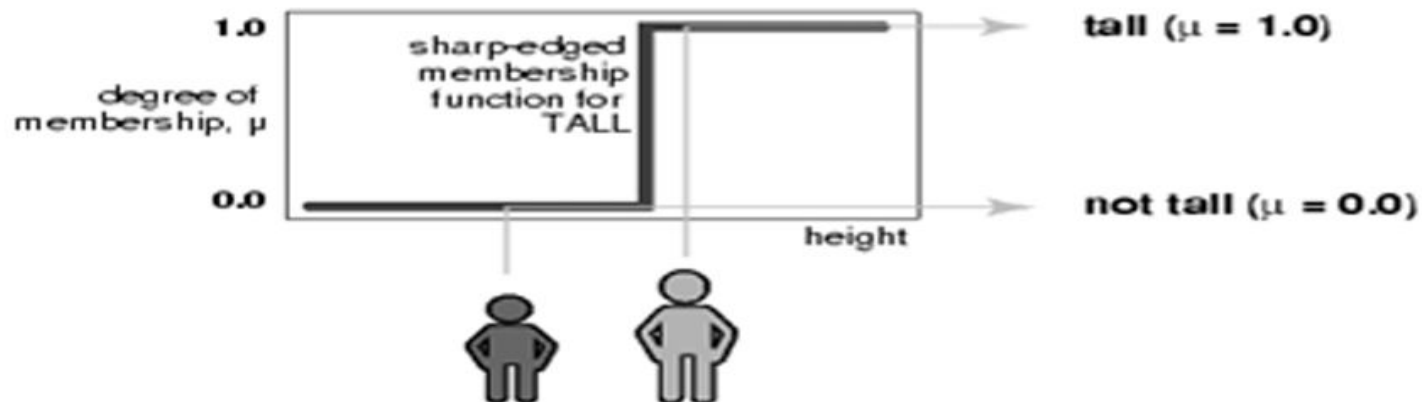


Fig.4.3: Illustration of crisp boundary of a set of heights of persons



Some Fundamental Definitions

Fuzzy Set: A fuzzy set is an extension of a crisp set. Crisp set allow only full membership or no membership at all, where as fuzzy sets allow partial membership.

Suppose X is a collection of objects and is collectively denoted by x [i.e., let $X(x)$ be a non-empty set], then a fuzzy set A in terms of X is defined as set of ordered pairs:

$$A = \{(x), \mu_A(x) : x \in X\}$$

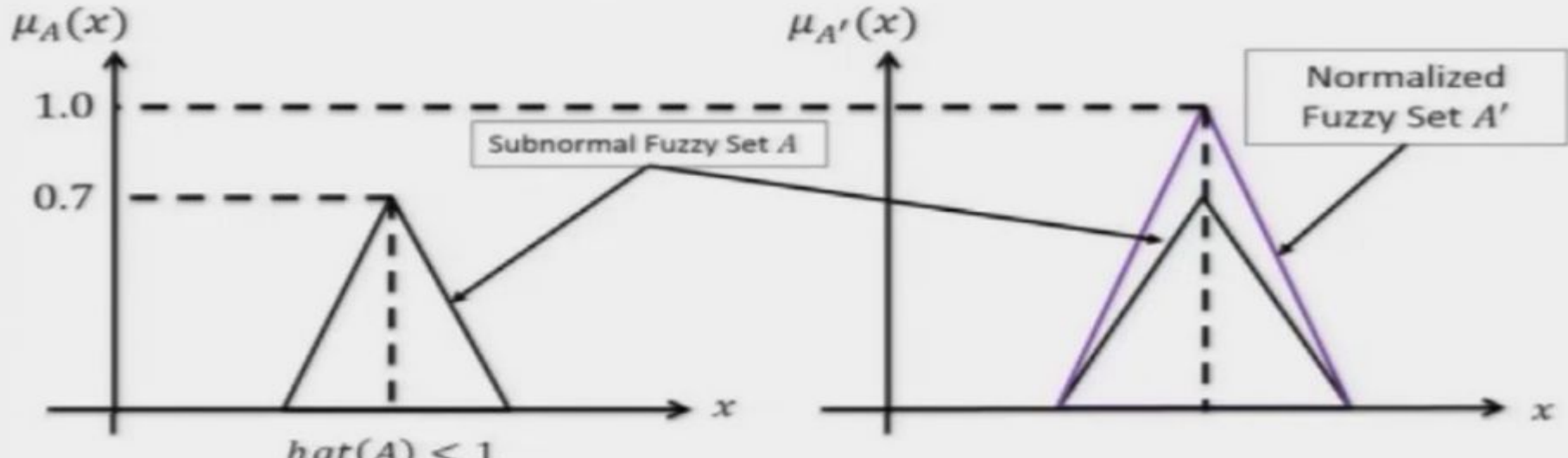
Where, $\mu_A(x)$ is called the membership function of grade of membership (also known as degree of truth or degree of compatibility).

$\mu(x)=1$ (**implies full membership**)

Normalized Fuzzy Set

The fuzzy set A' is the normalized version of a fuzzy set A which is represented by $A' = \text{Norm}(A)$. The normalization is carried out as follows:

$$A' = \text{Norm}(A) = \sum_x \frac{\mu_A(x)}{\text{hgt}(A)} / x$$



Example: A subnormal fuzzy set A is given for a universe of discourse X . Write down the normalized fuzzy set A' .

$$\underline{A} = \{(\underline{5,0.1}), (6,0.4), (7,0.5), (\underline{8,0.8}), (9,0.7), (10,0.3), (11,0.5), (12,0.7), (13,0.6), (14,0.5), (15,0.5)\}$$

Solution:

We have, $hgt(A) = 0.8$. The normalization process happens through the expression:

$$A' = Norm(A) = \sum_x \frac{\mu_A(x)}{hgt(A)} / x = \sum_x \frac{\mu_A(x)}{0.8} / x$$

Hence, the normalized fuzzy set A' will be given as:

$$\begin{aligned} &= \{(\underline{5,0.125}), (6,0.5), (\underline{7,0.625}), (\underline{8,1}), (9,0.875), (10,0.375), \\ & \quad (11,0.625), (12,0.875), (13,0.75), (14,0.625), (15,0.625)\} \end{aligned}$$

Support

The support of a fuzzy set A in X , is the crisp set of all elements $x \in X$, such that $\mu_A(x) > 0$. It is denoted by $S(A)$.

$$S(A) = \{x \in X: \mu_A(x) > 0\}$$

Crossover points

If A is a fuzzy set, then its crossover point is a point $x \in X$ at which $\mu_A(x) = 0.5$:

$$\text{Crossover}(A) = \{x | \mu_A(x) = 0.5\}$$

Example: Let the universe space $X = \{1, 2, 3, \dots, 15\}$ and

fuzzy set A of X be, $\{(2, 0), (3, 0.1), (4, 0.4), (5, 0.5), (7, 0.2), (8, 0.6), (9, 1.0), (10, 0), (11, 0)\}$.

Then, support of fuzzy $S(A) = \{3, 4, 5, 7, 8, 9\}$. and Crossover point of A is $x = 5$, since $\mu_A(x) = 0.5$.

Fuzzy Singleton

A fuzzy singleton is a fuzzy set that has a support of a single point in X and have membership function $\mu_A(x) = 1$.

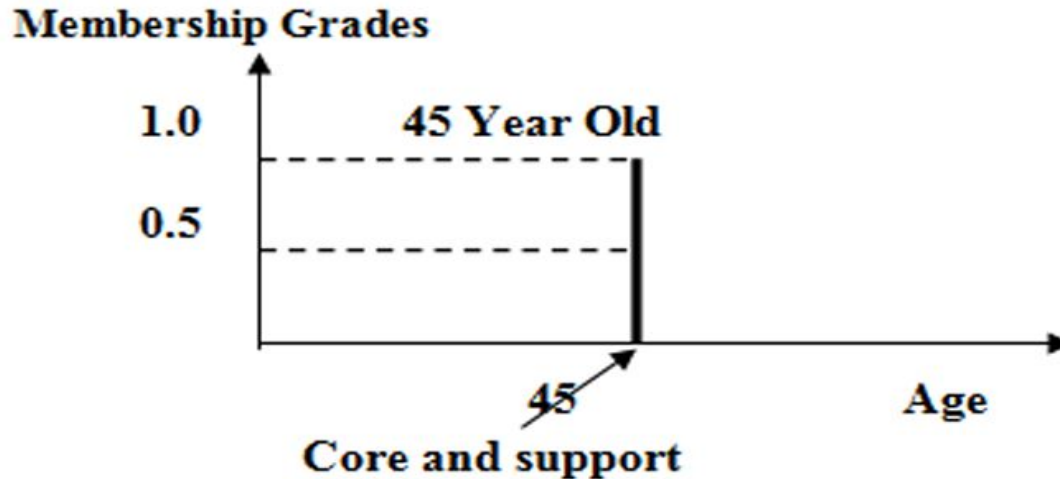
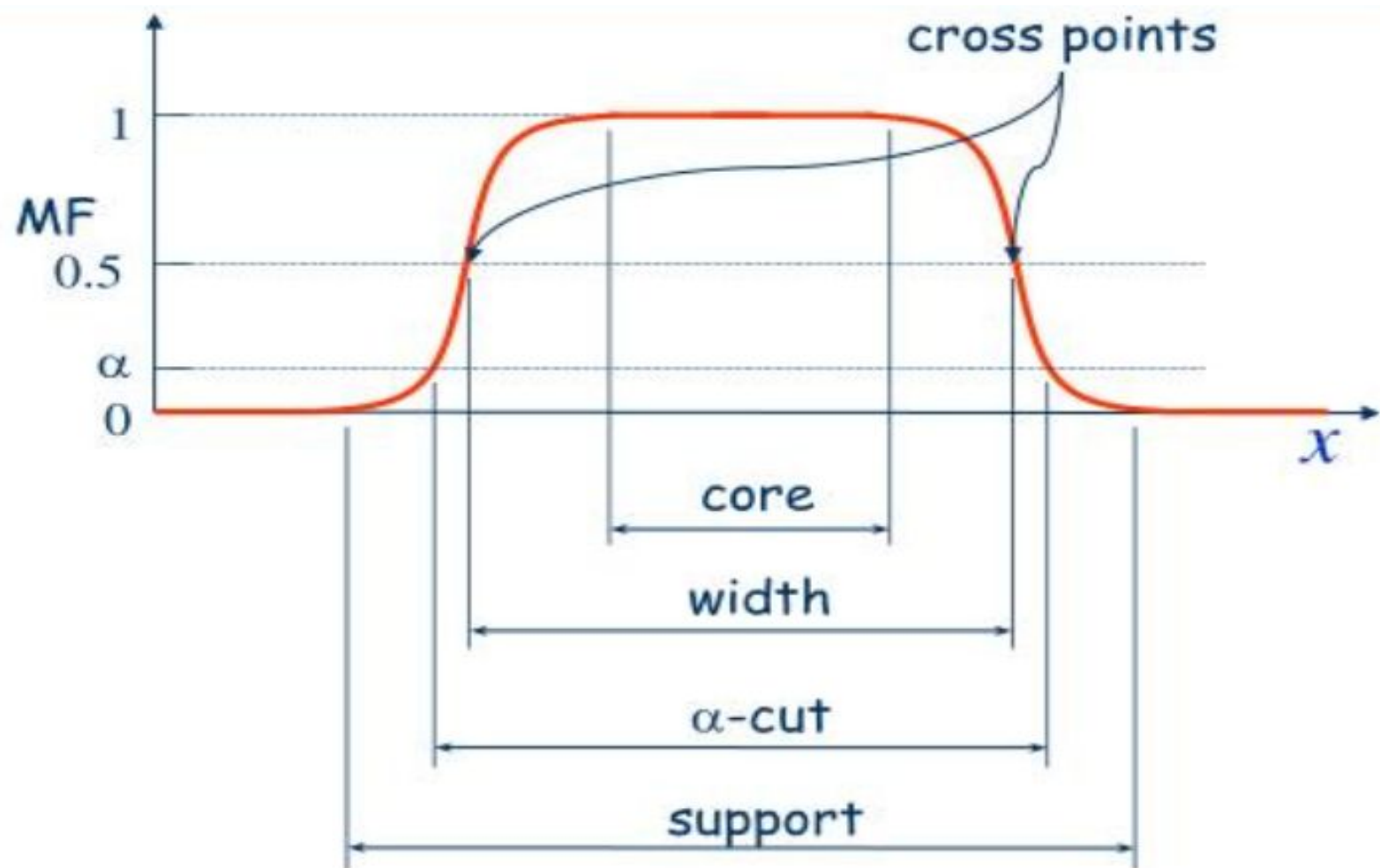


Fig.4.5: the fuzzy singleton "45 years old."

Core

In a fuzzy set A , the core is defined as, the set of all points x in X such that $\mu_A(x) = 1$:

$$\text{Core}(A) = \{x \mid \mu_A(x) = 1\}$$



α -cut, strong α -cut

The α -cut or α -level set of a fuzzy set A , is nothing but a crisp set defined by,

$$A_{\alpha} = \{x | \mu_A(x) \geq \alpha\}, \alpha \in [0,1]$$

An α -cut of a fuzzy set A is A_{α} that contains all the elements of X that have a membership greater than or equal to the specified value of α .

Strong α -cut or strong α -level set are defined similarly,

$$A_{\alpha}^i = \{x | \mu_A(x) > \alpha\}, \alpha \in [0, 1]$$

Example: A fuzzy set A is given for a universe of discourse X . Write down the α - Cut sets for $\alpha = 0.4, 0.6$ and Strong α - Cut sets for $\alpha = 0.6$.

$$\underline{A} = \{(5,0.1), \underline{(6,0.4)}, (7,0.5), (8,1), (9,0.7), (10,0.3), (11,1), (12,0.7), (13,0.6), (14,0.5), (15,0.5)\}$$

Solution:

- α - Cut for $\alpha = 0.4$, $\underline{\underline{A_{0.4}}} = \{\underline{6}, \underline{7}, \underline{8}, \underline{9}, \underline{11}, \underline{12}, \underline{13}, \underline{14}, \underline{15}\}$ ✓
- α - Cut for $\alpha = \underline{0.6}$, $\underline{\underline{A_{0.6}}} = \{\underline{8}, \underline{9}, \underline{11}, \underline{12}, \underline{13}\}$
- Strong α - Cut for $\alpha = 0.6$, $A'_{0.6} = \{8, 9, 11, 12\}$

Fuzzy set operations

Containment or subset:

Fuzzy set A is said to be the subset of B if and only if $\mu_A(x) \leq \mu_B(x)$ for all x.

Symbolically,

$$A \subseteq B \Leftrightarrow \mu_A(x) \leq \mu_B(x)$$

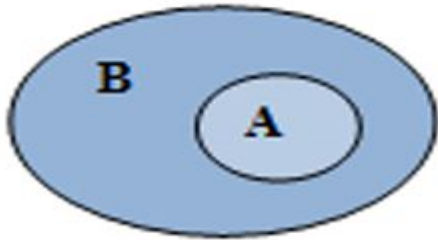


Fig.4.6: Concept of $A \subseteq B$

Union:

The union of two fuzzy sets A and B is related to each other by,

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

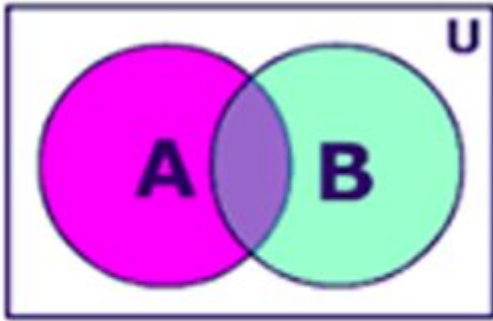


Fig.4.7: Concept of union.

Intersection

The intersection of two given fuzzy sets, A and B, is the “largest” fuzzy set which is contained in both A and B

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

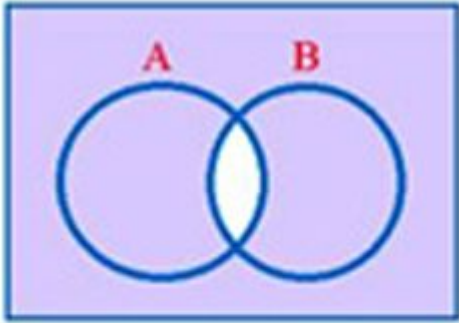


Fig.4.8: Concept of intersection

Complement (negation)

The complement of fuzzy set A, denoted by NOT A, is defined as,

$$\mu_{\text{not } A}(x) = (1 - \mu_A(x))$$

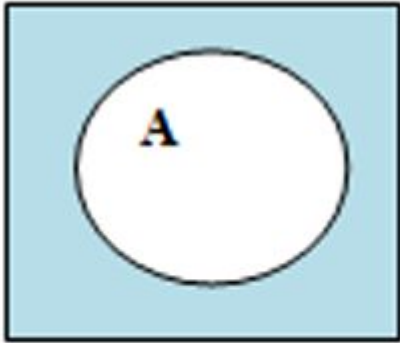


Fig.4.9: Concept of complement

Membership Function Formulation

A fuzzy set is completely characterized with its membership function.

A membership function (MF) is graphically represented by a curve that map each point in the input space to a membership value (or degree of membership) in the range between 0 and 1.

Membership function allow us to represent a fuzzy set graphically.

The x axis depicts the universe of discourse, whereas the y axis represents the grades of membership in the $[0, 1]$ interval.

There are various membership functions, the most commonly used ones are:
Triangular membership function

- Trapezoid membership function
- Gaussian membership function
- S-membership function
- Exponential membership functions
- Generalized bell shaped membership function
- Polynomial membership term

Triangular Membership Functions

Defined by a lower limit **a**, an upper limit **b**, and a value **m**, where $a < m < b$. the tuple $\{a, m, b\}$, denotes the location of the corresponding corners.

The triangular MF is given by:

$$\mu_A(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{m-a}, & a < x \leq m \\ \frac{b-x}{b-m}, & m < x < b \\ 0, & x \geq b \end{cases}$$

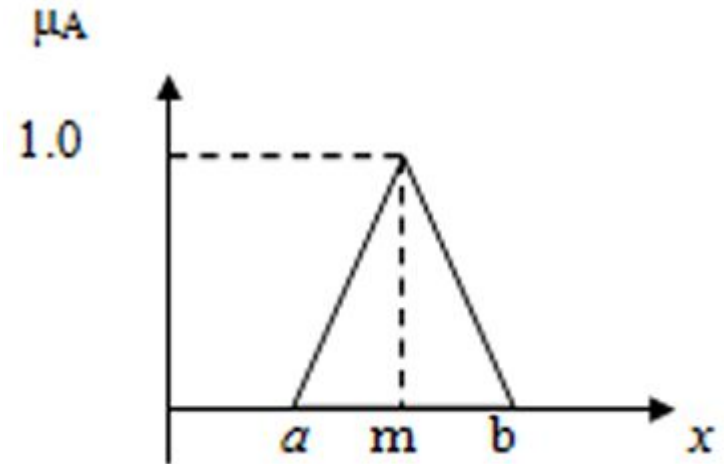


Fig. 4.10: Triangular curve

Trapezoidal Membership Function

Defined by a lower limit a , an upper limit d , a lower support limit b , and an upper support limit c , where $a < b < c < d$.

$$\mu_A(x) = \begin{cases} 0, & (x < a) \text{ or } (x > d) \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c \leq x \leq d \end{cases}$$

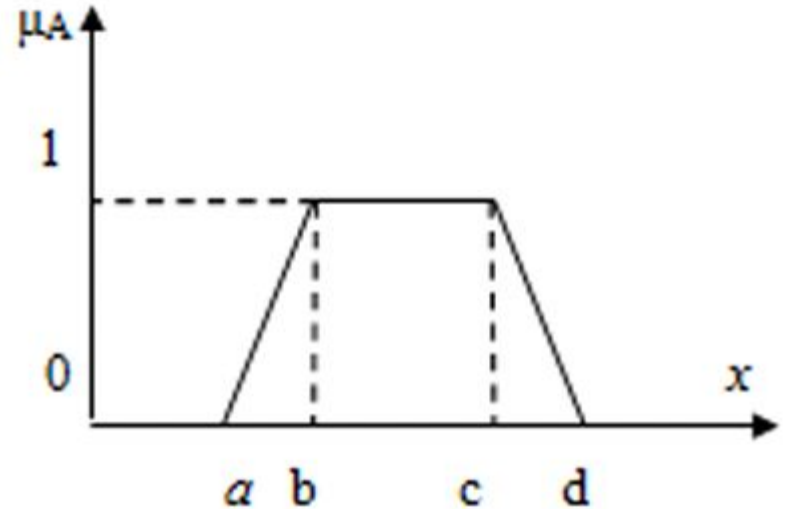


Fig.4.11: Trapezoidal Membership function

Two special categories of a trapezoidal function, called as R- function and L- function.

$$\mu_A(x) = \begin{cases} 0, & x > d \\ \frac{d-x}{d-c}, & c \leq x \leq d \\ 1, & x < c \end{cases}$$

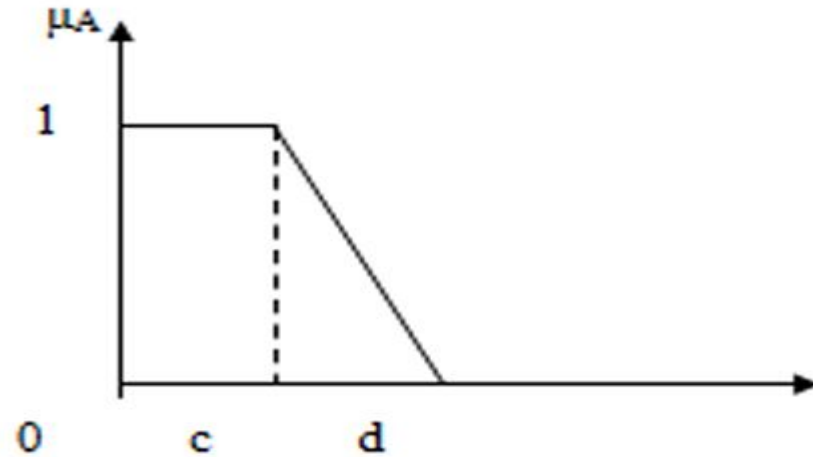


Fig.4.12: R- Membership Function

L-Functions

$$\mu_A(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & x > b \end{cases}$$

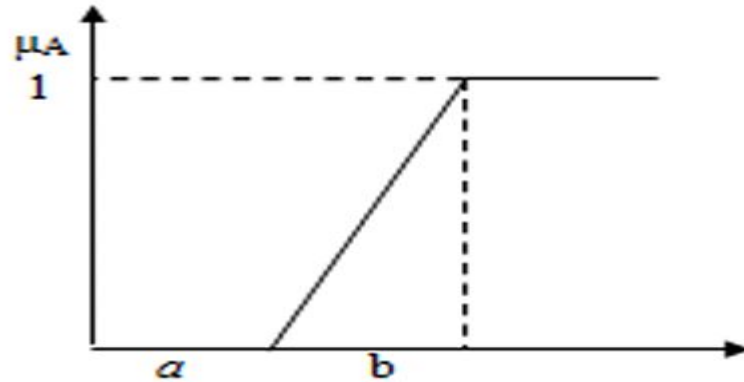


Fig.4.13: L-membership function

- Due to their simple formulae and computational efficiency, both triangular and trapezoidal MFs have been extensively used in real time applications.
- However, they are not smooth at the corner points.

Gaussian Membership Function

Described by a central parameter m and a standard deviation denoted by k ; $k > 0$. The smaller the value of k is, the narrower the “bell” is.

$$\mu_A(x) = e^{-\frac{(x-m)^2}{2k^2}}$$

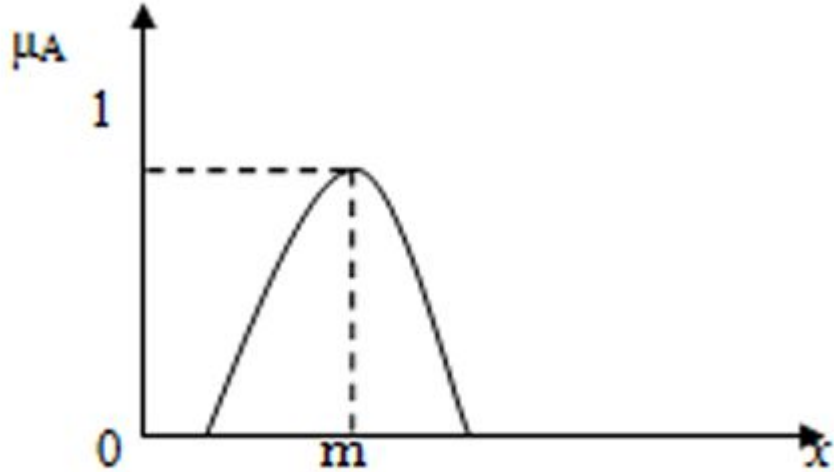


Fig.4.13: Gaussian membership function

S-membership function

$$\mu(x) = \begin{cases} 0, & \text{if } x \leq a \\ 2\left(\frac{x-a}{b-a}\right)^2, & \text{if } a \leq x \leq m \\ 1 - 2\left(\frac{x-b}{b-a}\right)^2, & \text{if } m \leq x \leq b \\ 1, & \text{if } x \geq b \end{cases}$$

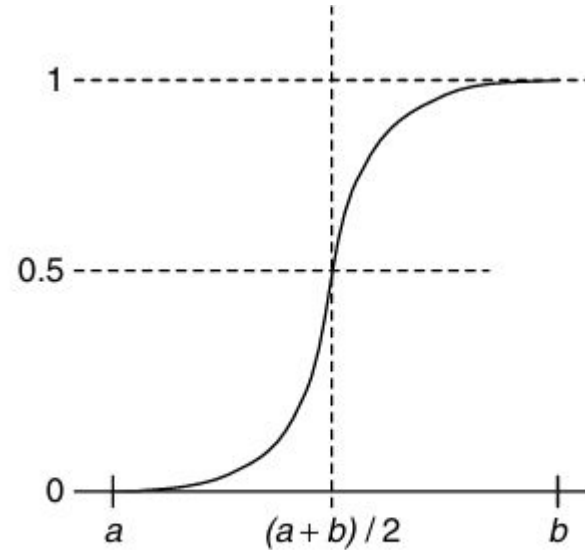


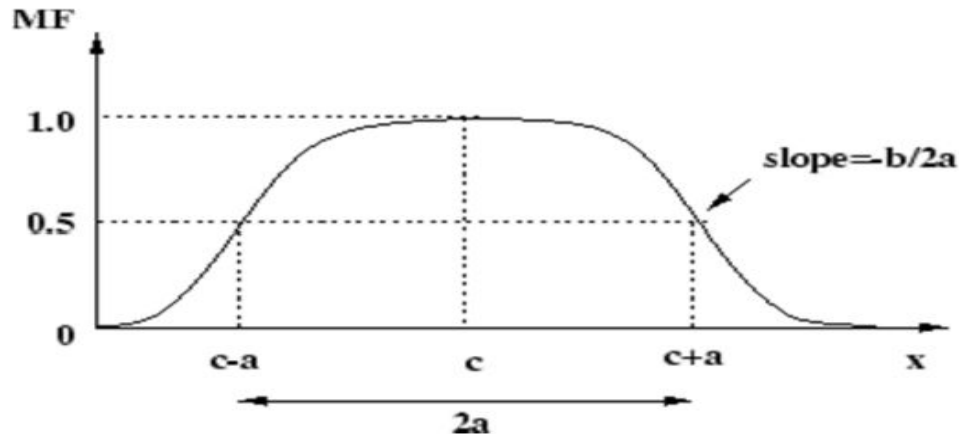
Fig. 2.12. The S - function

- The point $m = (a + b) / 2$ is known as the crossover point of the S-function.

Generalized Bell membership function

A generalized bell membership function has three parameters: a – responsible for its width, c – responsible for its center and b – responsible for its slopes.

$$gbellmf(x; a, b, c) = \frac{1}{1 + \left| \frac{x - c}{b} \right|^{2b}}$$



Fuzziness vs. Probability

Illustrative Example

A thirsty man in a desert is given two glasses of water:

- **Glass A (Fuzziness):** Water is **partially poisonous (0.1 degree)**—causes illness but not death.
- **Glass B (Probability):** Water is **either pure (90%) or fully poisonous (10%)**—could be fatal.

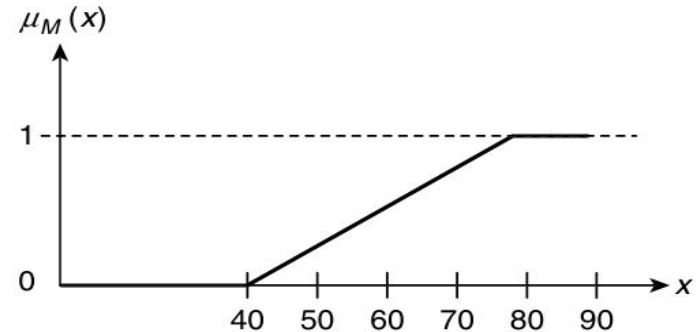
Key Difference

- **Fuzziness** represents **gradual change and partial truth** (shades of gray), where elements belong to a category to varying degrees.
- **Probability** represents **uncertainty and binary outcomes** (black or white), measuring how likely an event is to happen.

Fuzzy Membership

Let a, b, c, d, and e be five students who scored 55, 35, 60, 85 and 75 out of 100 respectively in Mathematics. The students constitute the universe of discourse $U = \{a, b, c, d, e\}$ and a fuzzy set M of the students who are good in Mathematics is defined on U with the help of the following membership function.

$$\mu_M(x) = \begin{cases} 0, & \text{if } x < 40 \\ \frac{x-40}{40}, & \text{if } 40 \leq x < 80 \\ 1, & \text{if } x \geq 80 \end{cases}$$



$M = \{(a, 0.375), (c, 0.5), (d, 1.0), (e, 0.875)\}$ OR

$$M = \frac{0.375}{a} + \frac{0.5}{c} + \frac{1.0}{d} + \frac{0.875}{e}$$

OPERATIONS ON FUZZY SETS

#	Operation/Relation	Description
1	Union ($P \cup Q$)	$\mu_{P \cup Q}(x) = \max \{\mu_P(x), \mu_Q(x)\}, \forall x \in U$
2	Intersection ($P \cap Q$)	$\mu_{P \cap Q}(x) = \min \{\mu_P(x), \mu_Q(x)\}, \forall x \in U$
3	Complementation (P')	$\mu_{P'}(x) = 1 - \mu_P(x), \forall x \in U$
4	Equality ($P = Q$)	Two fuzzy sets P and Q are equal if and only if $\forall x \in U, \mu_P(x) = \mu_Q(x)$
5	Inclusion ($P \subseteq Q$)	P is included in Q , i.e., P is a subset of Q , written as $P \subseteq Q$, if and only if $\forall x \in U, \mu_P(x) \leq \mu_Q(x)$
6	Product ($P \cdot Q$)	$\mu_{P \cdot Q}(x) = \mu_P(x) \times \mu_Q(x), \forall x \in U$
7	Difference ($P - Q$)	$P - Q = P \cap Q'$
8	Disjunctive sum ($P \oplus Q$)	$P \oplus Q = (P \cap Q') \cup (P' \cap Q)$

Example (Fuzzy Set Operation)

Table 2.1 *A family data set*

#	Family member	Age	Gender
1	Grand-pa	72	Male
2	Grand-ma	63	Female
3	Dad	41	Male
4	Mom	38	Female
5	Daughter	15	Female
6	Son	13	Male
7	Aunty	52	Female

U(universe of discourse): {Grand-pa, Grand-ma, Dad, Mom, Sister, Brother, Aunt}

Fuzzy set A of senior persons: $A = \{(\text{Grand-pa}, 1.0), (\text{Grand-ma}, 0.825), (\text{Dad}, 0.275), (\text{Mom}, 0.2), (\text{Aunty}, 0.55)\}$ OR

$$A = \frac{1.0}{\text{Grand-pa}} + \frac{0.825}{\text{Grand-ma}} + \frac{0.275}{\text{Dad}} + \frac{0.2}{\text{Mom}} + \frac{0.55}{\text{Aunty}}$$

The membership function for A is:

$$\mu_A(x) = \begin{cases} 0, & \text{if } \text{age}(x) \leq 30 \\ \frac{x-30}{40}, & \text{if } 30 \leq \text{age}(x) \leq 70 \\ 1, & \text{if } \text{age}(x) \geq 70 \end{cases}$$

Fuzzy set B of active persons:

$$\mu_B(x) = \begin{cases} 0, & \text{if } \text{age}(x) \leq 10 \\ \frac{x-10}{15}, & \text{if } 10 \leq \text{age}(x) \leq 25 \\ 1, & \text{if } 25 \leq \text{age}(x) \leq 50 \\ \frac{80-x}{30}, & \text{if } 50 \leq \text{age}(x) \leq 80 \\ 0, & \text{if } \text{age}(x) \geq 80 \end{cases}$$

$$B = \frac{0.267}{\text{Grand-pa}} + \frac{0.567}{\text{Grand-ma}} + \frac{1.0}{\text{Dad}} + \frac{1.0}{\text{Mom}} + \frac{0.333}{\text{Daughter}} + \frac{0.2}{\text{Son}} + \frac{0.933}{\text{Aunty}}$$

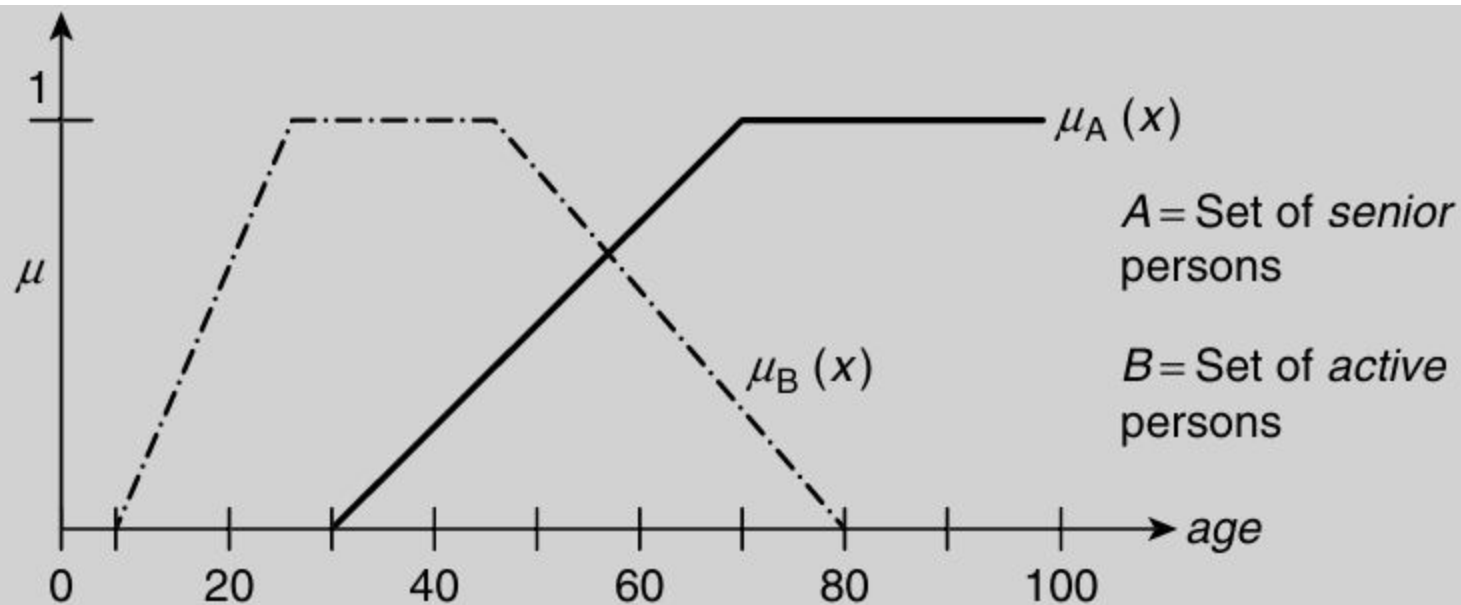


Fig. 2.24. Membership profiles of aged persons and active persons

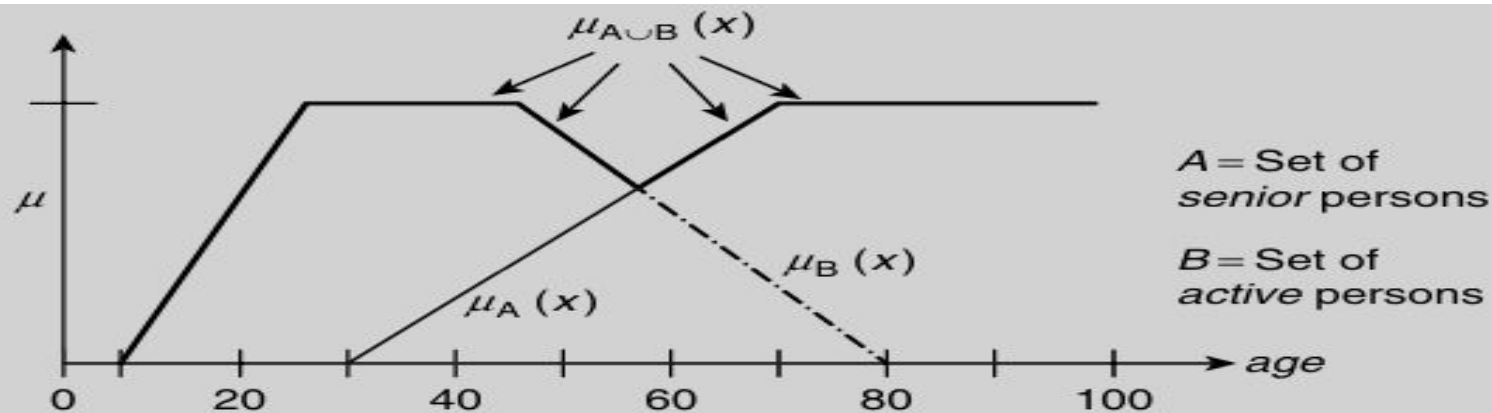


Fig. 2.25. Membership profile of senior OR active persons.

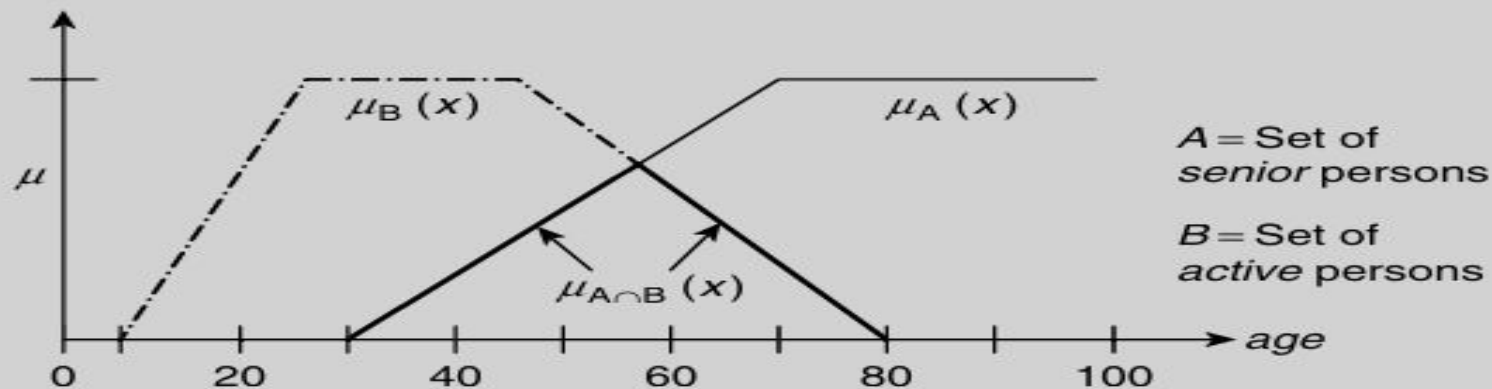


Fig. 2.26. Membership profiles of senior AND active persons.

Table 2.4. Operations on Fuzzy Sets Senior and Active Persons

#	Operation	Resultant fuzzy set
1.	$A \cup B$ senior OR active	$\{(Grand-pa, 1.0), (Grand-ma, 0.825), (Dad, 1.0), (Mom, 1.0), (Daughter, 0.333), (Son, 0.2), (Aunty, 0.933)\}$
2.	$A \cap B$ senior AND Active	$\{(Grand-pa, 0.267), (Grand-ma, 0.567), (Dad, 0.275), (Mom, 0.2), (Aunty, 0.55)\}$
3.	A', B' NOT senior, NOT active	$A' = \{(Grand-ma, 0.175), (Dad, 0.725), (Mom, 0.8), (Sister, 1.0), (Brother, 1.0), (Aunty, 0.45)\}$ $B' = \{(Grand-pa, 0.733), (Grand-ma, 0.433), (Sister, 0.667), (Brother, 0.8), (Aunty, 0.067)\}$

Example 2

Consider two given fuzzy sets

$$\underline{A} = \left\{ \frac{1}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0.2}{8} \right\}$$

$$\underline{B} = \left\{ \frac{0.5}{2} + \frac{0.4}{4} + \frac{0.1}{6} + \frac{1}{8} \right\}$$

Perform union, intersection, difference and complement over fuzzy sets $\underline{A}$ and $\underline{B}$.

Solution: For the given fuzzy sets we have the following

(a) Union

$$\begin{aligned}\underline{A} \cup \underline{B} &= \max\{\mu_{\underline{A}}(x), \mu_{\underline{B}}(x)\} \\ &= \left\{ \frac{1}{2} + \frac{0.4}{4} + \frac{0.5}{6} + \frac{1}{8} \right\}\end{aligned}$$

(b) Intersection

$$\begin{aligned}\underline{A} \cap \underline{B} &= \min\{\mu_{\underline{A}}(x), \mu_{\underline{B}}(x)\} \\ &= \left\{ \frac{0.5}{2} + \frac{0.3}{4} + \frac{0.1}{6} + \frac{0.2}{8} \right\}\end{aligned}$$

(c) Complement

$$\begin{aligned}\bar{\underline{A}} &= 1 - \mu_{\underline{A}}(x) = \left\{ \frac{0}{2} + \frac{0.7}{4} + \frac{0.5}{6} + \frac{0.8}{8} \right\} \\ \bar{\underline{B}} &= 1 - \mu_{\underline{B}}(x) = \left\{ \frac{0.5}{2} + \frac{0.6}{4} + \frac{0.9}{6} + \frac{0}{8} \right\}\end{aligned}$$

(d) Difference

$$\begin{aligned}\underline{A} \setminus \underline{B} &= \underline{A} \cap \bar{\underline{B}} = \left\{ \frac{0.5}{2} + \frac{0.3}{4} + \frac{0.5}{6} + \frac{0}{8} \right\} \\ \underline{B} \setminus \underline{A} &= \underline{B} \cap \bar{\underline{A}} = \left\{ \frac{0}{2} + \frac{0.4}{4} + \frac{0.1}{6} + \frac{0.8}{8} \right\}\end{aligned}$$

Properties of fuzzy set operations

#	Law	Description
1	Associativity	(a) $(P \cup Q) \cup R = P \cup (Q \cup R)$ (b) $(P \cap Q) \cap R = P \cap (Q \cap R)$
2	Commutativity	(a) $P \cup Q = Q \cup P$ (b) $P \cap Q = Q \cap P$
3	Distributivity	(a) $P \cup (Q \cap R) = (P \cup Q) \cap (P \cup R)$ (b) $P \cap (Q \cup R) = (P \cap Q) \cup (P \cap R)$
4	Idempotency	(a) $P \cup P = P$ (b) $P \cap P = P$
5	De Morgan's law	(a) $(P \cup Q)' = P' \cap Q'$ (b) $(P \cap Q)' = P' \cup Q'$
6	Boundary Conditions	(a) $P \cup \Phi = P, P \cup U = U$ (b) $P \cap \Phi = \Phi, P \cap U = P$
7	Involution	$(P')' = P$
8	Transitivity	If $P \subseteq Q$ and $Q \subseteq R$ then $P \subseteq R$

Note: The relations $P \cup P' = U$, and $P \cap P' = \Phi$ obeyed by crisp sets are no longer valid for fuzzy sets.

FUZZY RELATIONS

- Fuzzy sets as generalization of crisp sets where the degree of inclusiveness of an element may be anything from 0 to 1.
- In a similar fashion, the concept of a relation between two, or more, sets can be generalized to fuzzy relation.

Crisp Relations:

- Crisp relation R between two crisp sets A and B , two elements, x from A and y from B , are either related, or not. **(no scope of being partially related)**
- Crisp relation is defined as a subset of the Cartesian product of sets

Example 2.13 (*Cartesian product*)

Let $A = \{p, q, r\}$, and $B = \{1, 2, 3\}$. Then $A \times B = \{(p, 1), (p, 2), (p, 3), (q, 1), (q, 2), (q, 3), (r, 1), (r, 2), (r, 3)\}$, and $B \times A = \{(1, p), (1, q), (1, r), (2, p), (2, q), (2, r), (3, p), (3, q), (3, r)\}$.

Definition 2.20 (*Crisp relation*) Given two crisp sets A and B , a crisp relation R between A and B is a subset of $A \times B$.

$$R \subseteq A \times B$$

Example 2.15 (*Crisp relation*)

Consider the sets $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$ and the relation $R = \{(a, b) \mid b = a + 1, a \in A, \text{ and } b \in B\}$. Then $R = \{(1, 2), (2, 3), (3, 4)\}$. Obviously, here $R \subset A \times B$.

Example

Consider the elements defined in the universes X and Y as follows:

$$X = \{2, 4, 6\}; \quad Y = \{p, q, r\}$$

The Cartesian product of these two sets leads to

$$X \times Y = \{(p, 2), (p, 4), (p, 6), (q, 2), (q, 4), (q, 6), (r, 2), (r, 4), (r, 6)\}$$

From this set one may select a subset such that

$$R = \{(p, 2), (q, 4), (r, 4), (r, 6)\}$$

In relation matrix form:

R	p	q	r
2	1	0	0
4	0	1	1
6	0	0	1

The entries of relation matrix are defined as:

$$T_{ij} = \begin{cases} 1, & \text{if } (a_i, b_j) \in R \\ 0, & \text{otherwise} \end{cases}$$

Example:

Let $P = \{\text{Tony, Bobby, Mike}\}$ be a set of three students and $Q = \{\text{Math, Phy, Chem}\}$ be a set of three subjects in which Tony, Bobby, and Mike have taken a test.

Table 2.6. *Grades obtained by three students*

	<i>Math</i>	<i>Phy</i>	<i>Chem</i>
<i>Tony</i>	<i>C</i>	<i>B</i>	<i>A</i>
<i>Bobby</i>	<i>A</i>	<i>A</i>	<i>B</i>
<i>Mike</i>	<i>C</i>	<i>A</i>	<i>A</i>

A relation R between a student and a subject in which they have secured A grade as:

$$R = \{(x, y) \mid x \in P, y \in Q, \text{ and } x \text{ has secured grade A in subject } y\}.$$

The corresponding relation matrix is:

$$T_R = \begin{matrix} & \begin{matrix} \textit{Math} & \textit{phy} & \textit{Chem} \end{matrix} \\ \begin{matrix} T \\ B \\ M \end{matrix} & \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{matrix}$$

Operations on relations

Union, intersection, complementation, composition.

Example:

Let relation matrices T_R , T_S , T_T are:

$$T_R = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \end{matrix} \quad T_S = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix} \quad T_T = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$T_{R \cup S} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix} \quad T_{R \cap S} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix} \quad T_{R'} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Operations on relations (cont.)

Composition ($R \circ S$):

$(0, 1) \in R \circ T$ because $(0, 2) \in R$ and $(2, 1) \in T$.

$(0, 2) \in R$	and	$(2, 1) \in T$,	$\therefore (0, 1) \in R \circ T$
$(0, 2) \in R$	and	$(2, 3) \in T$,	$\therefore (0, 3) \in R \circ T$
$(1, 1) \in R$	and	$(1, 0) \in T$,	$\therefore (1, 0) \in R \circ T$
$(1, 1) \in R$	and	$(1, 2) \in T$,	$\therefore (1, 2) \in R \circ T$
$(2, 0) \in R$	and	$(0, 1) \in T$,	$\therefore (2, 1) \in R \circ T$
$(2, 2) \in R$	and	$(2, 3) \in T$,	$\therefore (2, 3) \in R \circ T$
$(3, 1) \in R$	and	$(1, 0) \in T$,	$\therefore (3, 0) \in R \circ T$
$(3, 1) \in R$	and	$(1, 2) \in T$,	$\therefore (3, 2) \in R \circ T$

Hence the relation matrix $T_{R \circ T}$ for $R \circ T$ looks like

$$T_{R \circ T} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Fuzzy Relations

- Fuzzy relations: Generalization of crisp relations.
- Entries of the relation matrix would be anything between 0 and 1.

Example 2.19 (Fuzzy relation)

Let $P = \{\text{money}, \text{fame}, \text{power}\}$ and $Q = \{\text{politics}, \text{showbiz}, \text{academics}\}$ be two crisp sets. Then R might be an imaginary relation between P and Q expressed by the relation matrix

$$T_R = \begin{matrix} & \begin{matrix} ps & sz & as \end{matrix} \\ \begin{matrix} m \\ f \\ p \end{matrix} & \begin{bmatrix} 0.7 & 0.9 & 0.6 \\ 0.8 & 0.9 & 0.5 \\ 1.0 & 0.7 & 0.3 \end{bmatrix} \end{matrix} \quad \begin{matrix} m : \text{money} & ps : \text{politics} \\ f : \text{fame} & sz : \text{showbiz} \\ p : \text{power} & as : \text{academics} \end{matrix}$$

Fuzzy Cartesian product:

- Let A and B be two fuzzy sets on reference sets X and Y respectively. The fuzzy Cartesian product of A and B, denoted by $A \times B$, is defined as $A \times B \subseteq X \times Y$, and $\mu_{A \times B}(a, b) = \min \{\mu_A(a), \mu_B(b)\} \quad \forall a \in X, \forall b \in Y$.

Example: Fuzzy sets A and B:

$$A = \frac{0.3}{m} + \frac{0.7}{n}, \quad B = \frac{0.5}{p} + \frac{0.1}{q} + \frac{0.8}{r}$$

$$A \times B = \begin{matrix} & p & q & r \\ \begin{matrix} m \\ n \end{matrix} & \begin{bmatrix} 0.3 & 0.1 & 0.3 \\ 0.5 & 0.1 & 0.7 \end{bmatrix} \end{matrix}$$

- A fuzzy relation R between two fuzzy sets A and B is a subset of the fuzzy Cartesian product $A \times B$. Hence $R \subseteq A \times B$.

Operations on Fuzzy Relations

Union	$(R \cup S)(x, y) = \max \{R(x, y), S(x, y)\}$
Intersection	$(R \cap S)(x, y) = \min \{R(x, y), S(x, y)\}$
Complementation	$R'(x, y) = 1 - R(x, y)$
Inclusion	$R \subseteq S$ if $\forall x, y, R(x, y) \leq S(x, y)$
Dominance	$R \supseteq S$ if $\forall x, y, R(x, y) \geq S(x, y)$
Equality	$R = S$ if $\forall x, y, R(x, y) = S(x, y)$

Example:

$$P = \text{Beautiful flowers} = \frac{0.3}{\text{Jasmine}} + \frac{0.9}{\text{Rose}} + \frac{1.0}{\text{Lotus}} + \frac{0.7}{\text{Daffodil}} + \frac{0.5}{\text{Sunflower}} + \frac{0.4}{\text{Hibiscus}} + \frac{0.6}{\text{Chrysanthemum}}$$

$$Q = \text{Fragrant flowers} = \frac{1.0}{\text{Jasmine}} + \frac{1.0}{\text{Rose}} + \frac{0.5}{\text{Lotus}} + \frac{0.2}{\text{Daffodil}} + \frac{0.2}{\text{Sunflower}} + \frac{0.1}{\text{Hibiscus}} + \frac{0.4}{\text{Chrysanthemum}}$$

Compute the fuzzy sets $P \cup Q$, $P \cap Q$, P' , Q' , $P - Q$, $P \oplus Q$. Also, verify that $P \cup P' \neq U$, $P \cap P' \neq \phi$.

$$P \cup Q = \frac{1.0}{\text{Jasmine}} + \frac{1.0}{\text{Rose}} + \frac{1.0}{\text{Lotus}} + \frac{0.7}{\text{Daffodil}} + \frac{0.5}{\text{Sunflower}} + \frac{0.4}{\text{Hibiscus}} + \frac{0.6}{\text{Chrysanthemum}}$$

$$P \cap Q = \frac{0.3}{\text{Jasmine}} + \frac{0.9}{\text{Rose}} + \frac{0.5}{\text{Lotus}} + \frac{0.2}{\text{Daffodil}} + \frac{0.2}{\text{Sunflower}} + \frac{0.1}{\text{Hibiscus}} + \frac{0.4}{\text{Chrysanthemum}}$$

$$\begin{aligned}
 P' &= \frac{0.7}{Jasmine} + \frac{0.1}{Rose} + \frac{0}{Lotus} + \frac{0.3}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.6}{Hibiscus} + \frac{0.4}{Chrysanthemum} \\
 &= \frac{0.7}{Jasmine} + \frac{0.1}{Rose} + \frac{0.3}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.6}{Hibiscus} + \frac{0.4}{Chrysanthemum}
 \end{aligned}$$

$$\begin{aligned}
 P - Q &= P \cap Q' \\
 &= \frac{0}{Jasmine} + \frac{0}{Rose} + \frac{0.5}{Lotus} + \frac{0.7}{Daffodil} + \frac{0.5}{Sunflower} + \frac{0.4}{Hibiscus} + \frac{0.6}{Chrysanthemum}
 \end{aligned}$$

$$P \oplus Q = (P \cap Q') \cup (P' \cap Q)$$

FUZZY LOGIC BASICS

Fuzzy Truth Values:

- In classical (or crisp) logic there are only two possible truth values, true and false.
- Unlike crisp truth values, there are various fuzzy truth values including the crisp truth values.

Table 3.18. *Fuzzy truth values*

Linguistic	Numeric (tentative)
Absolutely False	0.00
Partly False	0.25
Neither False nor True	0.50
Both False and True	0.50
Partly True	0.75
Absolutely True	1.00

FUZZY LOGIC BASICS

Fuzzy Truth Values:

- In classical (or crisp) logic there are only two possible truth values, true and false.
- Unlike crisp truth values, there are various fuzzy truth values including the crisp truth values.

Example:

- **Classical Logic:** A temperature sensor defines “Hot” as temperatures **above 30°C** (strict boundary).
- **Fuzzy Logic:** Instead of a hard threshold, we define a **membership function** where:
 - 28°C → 0.2 (Slightly Hot)
 - 30°C → 0.5 (Moderately Hot)
 - 35°C → 0.9 (Very Hot)
 - 40°C → 1 (Completely Hot)

This enables smooth transitions rather than abrupt changes.

Fuzzy Proposition

A **fuzzy proposition** is a statement where truth values are **not strictly true or false** but lie in a range **[0,1]**, indicating the degree of truth.

Example:

- Classical Proposition: "The water is hot" (either **true or false**).
- Fuzzy Proposition: "The water is **somewhat hot** (0.6 truth value)" or "The water is **very hot** (0.9 truth value)".

This approach helps in natural decision-making where conditions are subjective.

Linguistic Variables: Expressing Propositions in Human Terms

A **linguistic variable** takes **words instead of numbers** to describe a condition. These words are represented by **fuzzy sets**.

Example:

For **Speed of a Car**, instead of exact numbers, we use:

- **Slow** (0–40 km/h)
- **Moderate** (30–70 km/h)
- **Fast** (60–100 km/h)

Fuzzy Logic Operations: Making Decisions Based on Fuzzy Propositions

Fuzzy logic uses special operations to process fuzzy values. These are similar to classical logic operations but work with **degrees of truth**.

Basic Operations:

1. **Fuzzy Union (OR, max)** – "At least one condition should hold."
 - If "Weather is Warm" = 0.6
 - If "Weather is Humid" = 0.8
 - Then **"Weather is Warm OR Humid" = $\max(0.6, 0.8) = 0.8$**
2. **Fuzzy Intersection (AND, min)** – "Both conditions should hold."
 - If "Traffic is Heavy" = 0.7
 - If "Road is Slippery" = 0.5
 - Then **"Traffic is Heavy AND Road is Slippery" = $\min(0.7, 0.5) = 0.5$**

3. Fuzzy Negation (NOT, 1 - value) – "Opposite of a condition."

- If "Air is Polluted" = 0.6
- Then **"Air is NOT Polluted" = 1 - 0.6 = 0.4**

Table 3.20. *Fuzzy logic operations*

#	Operation	Symbol	Usage	Definition
1	<i>disjunction</i>	+	$P + Q$	$\tau(P + Q) = \max \{ \tau(P), \tau(Q) \}$
2	<i>conjunction</i>	•	$P \bullet Q$	$\tau(P \bullet Q) = \min \{ \tau(P), \tau(Q) \}$
3	<i>negation</i>	$\neg$	$\neg P$	$\tau(\neg P) = 1 - \tau(P)$
4	<i>implication</i>	$\rightarrow$	$P \rightarrow Q$	$\tau(P \rightarrow Q) = \max \{ 1 - \tau(P), \tau(Q) \}$

Example 3.16*(Fuzzy logic operations)*

Let us consider the following two fuzzy propositions along with their fuzzy truth values

p = Mr. Bakshi is handsome. $\tau(p) = 0.7$

q = Mr. Bakshi is tall. $\tau(q) = 0.4$

Various fuzzy operations on these fuzzy operations are shown below.

Fuzzy AND. $\tau(p \bullet q) = \min \{ \tau(p), \tau(q) \} = \min \{ 0.7, 0.4 \} = 0.4$

Fuzzy OR. $\tau(p + q) = \max \{ \tau(p), \tau(q) \} = \max \{ 0.7, 0.4 \} = 0.7$

Fuzzy NOT. $\tau(p') = 1 - \tau(p) = 1 - 0.7 = 0.3,$

$\tau(q') = 1 - \tau(q) = 1 - 0.4 = 0.6$

Fuzzy implication $\tau(p \rightarrow q) = \max \{ 1 - \tau(p), \tau(q) \} = \max \{ 1 - 0.7, 0.4 \} = 0.4$

Fuzzy truth and linguistic variables

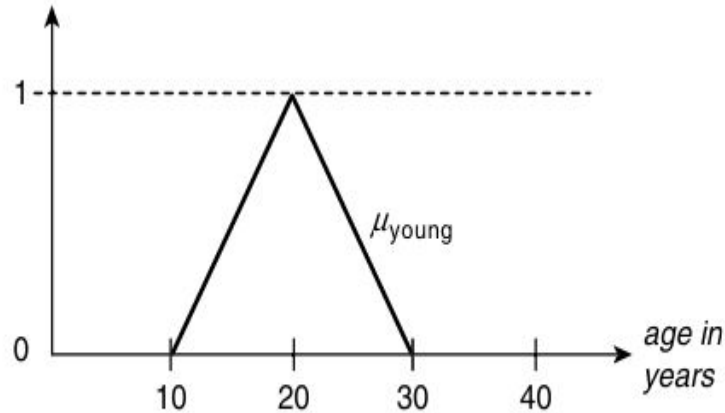


Fig. 3.1. Membership profile of young

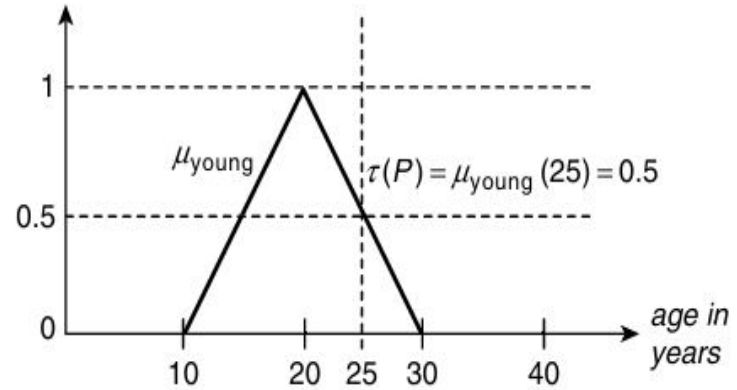


Fig. 3.2. Finding fuzzy truth value from degree of membership

- Consider the fuzzy proposition $p = \text{'Anita is young'}$.
- predicate young is a linguistic value of the linguistic variable age.
- Other possible linguistic values of age are very young, middle-aged, aged, old, very old, and so on.
- Each of these linguistic values corresponds to a fuzzy set on the universe of age.

FUZZY RULES

- The core of fuzzy inference system is the fuzzy rule base, which consists of a set of fuzzy rules.
- An elementary fuzzy rule R is a statement of the form:

If p Then q

- p and q are atomic fuzzy propositions known as the antecedent and the consequent respectively.
- A fuzzy statement p = 'x is A': Its truth value is given by the membership value of the fuzzy set A
- The fuzzy rule R : **If 'x is A' Then 'y is B'**
- Symbolically expressed as $R : \mathbf{A(x) \rightarrow B(y)}$

The fuzzy rule **If 'x is A' Then 'y is B'** can be expressed as a fuzzy relation between A and B where

$$R(x, y) = \tau [A(x) \rightarrow B(y)]$$

Two popular interpretations of fuzzy rule are by:

1. Mamdani
2. Zadeh

- Mamdani's interpretation of fuzzy rule:

$$\begin{aligned} R(x, y) &= \tau [A(x) \rightarrow B(y)] \\ &= \min [\tau(A(x)), \tau(B(y))] = \min [\mu_A(x), \mu_B(y)] \end{aligned}$$

- Zadeh's interpretation of fuzzy rule:

$$\begin{aligned} R(x, y) &= \tau [A(x) \rightarrow B(y)] \\ &= \max [\min \{\mu_A(x), \mu_B(y)\}, 1 - \mu_A(x)] \end{aligned}$$

- Equivalent to the relation:

$$R = (A \times B) \cup (A' \times V)$$

Example

Let R : If 'job is risky' Then 'compensation is high' be a fuzzy rule. There are four jobs job_1, job_2, job_3 and job_4 , constituting the universe $job = \{job_1, job_2, job_3, job_4\}$. Also, there are four categories of compensation c_1, c_2, c_3 , and c_4 in ascending order. Hence the universe for compensations is $compensation = \{c_1, c_2, c_3, c_4\}$. The fuzzy sets *risky-job* and *high-compensation* are defined on the universes *job* and *compensation* respectively as given below.

$$risky-job = \frac{0.3}{job_1} + \frac{0.8}{job_2} + \frac{0.7}{job_3} + \frac{0.9}{job_4}$$
$$high-compensation = \frac{0.2}{c_1} + \frac{0.4}{c_2} + \frac{0.6}{c_3} + \frac{0.8}{c_4}$$

Using Zadeh's interpretation, the truth value of rule R is expressed by the relation:

$$R = (\text{risky-job} \times \text{high-compensation}) \cup (\overline{\text{risky-job}} \times \text{compensation})$$

Now,

$$\text{risky-job} \times \text{high-compensation} = \begin{matrix} & C_1 & C_2 & C_3 & C_4 \\ \begin{matrix} job_1 \\ job_2 \\ job_3 \\ job_4 \end{matrix} & \begin{bmatrix} 0.2 & 0.3 & 0.3 & 0.3 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.2 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \end{bmatrix} \end{matrix},$$

$$\text{and, } \overline{\text{risky-job}} \times \text{compensation} = \begin{matrix} & C_1 & C_2 & C_3 & C_4 \\ \begin{matrix} job_1 \\ job_2 \\ job_3 \\ job_4 \end{matrix} & \begin{bmatrix} 0.7 & 0.7 & 0.7 & 0.7 \\ 0.2 & 0.2 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \end{bmatrix} \end{matrix},$$

and finally, $R = (\text{risky-job} \times \text{high-compensation}) \cup (\overline{\text{risky-job}} \times \text{compensation})$

$$= \begin{matrix} job_1 \\ job_2 \\ job_3 \\ job_4 \end{matrix} \begin{bmatrix} 0.7 & 0.7 & 0.7 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \\ 0.3 & 0.4 & 0.6 & 0.7 \\ 0.2 & 0.4 & 0.6 & 0.8 \end{bmatrix}$$

Fuzzy If-Then-Else

- A fuzzy If-Then-Else rule R has the form: $\text{If 'x is A' Then 'y is B' Else 'y is C'}$

where A is a fuzzy set on some universe U and B and C are fuzzy sets on the universe V.

- The truth value of R:

$$\begin{aligned}\tau(R(x, y)) &= \mu_R(x, y) \\ &= \max [\min \{\mu_A(x), \mu_B(y)\}, \min \{1 - \mu_A(x), \mu_C(y)\}]\end{aligned}$$

- Represented as:

$$R = (A \times B) \cup (A' \times C)$$

Example

Example 3.20

(Fuzzy If-Then-Else rule)

The fuzzy If-Then-Else rule under consideration is R : If '*distance is long*' Then '*drive at high speed*' Else '*drive at moderate speed*'. To match with the form given in Expression 3.8, this rule can be restated as R : If '*distance is long*' Then '*speed is high*' Else '*speed is moderate*'. The relevant sets (crisp and fuzzy) are, $distance = \{100, 500, 1000, 5000\}$ is the universe of the fuzzy set *long-distance*, $speed = \{30, 50, 70, 90, 120\}$ is the universe of the fuzzy sets *high-speed* as well as *moderate-speed*, and

$$long-distance = \frac{0.1}{100} + \frac{0.3}{500} + \frac{0.7}{1000} + \frac{1.0}{5000}$$

$$high-speed = \frac{0.1}{30} + \frac{0.3}{50} + \frac{0.5}{70} + \frac{0.7}{90} + \frac{0.9}{120}$$

$$moderate-speed = \frac{0.3}{30} + \frac{0.8}{50} + \frac{0.6}{70} + \frac{0.4}{90} + \frac{0.1}{120}$$

$$R = (A \times B) \cup (A' \times C)$$

$$= (\text{long-distance} \times \text{high-speed}) \cup (\overline{\text{long-distance}} \times \text{normal-speed}).$$

The relation matrix R is computed as follows.

$$\text{long-distance} \times \text{high-speed} =$$

	30	50	70	90	120
100	0.1	0.1	0.1	0.1	0.1
500	0.1	0.3	0.3	0.3	0.3
1000	0.1	0.3	0.5	0.7	0.7
5000	0.1	0.3	0.5	0.7	0.9

$$\overline{\text{long-distance}} \times \text{moderate-speed} =$$

	30	50	70	90	120
100	0.3	0.8	0.6	0.4	0.9
500	0.3	0.7	0.6	0.4	0.1
1000	0.3	0.3	0.3	0.3	0.1
5000	0.0	0.0	0.0	0.0	0.0

$$\therefore R = (A \times B) \cup (A' \times C)$$

$$= (\text{long-distance} \times \text{high-speed}) \cup (\overline{\text{long-distance}} \times \text{normal-speed})$$

	30	50	70	90	120
100	0.3	0.8	0.6	0.4	0.9
500	0.3	0.7	0.6	0.4	0.3
1000	0.3	0.3	0.5	0.7	0.7
5000	0.1	0.3	0.5	0.7	0.9